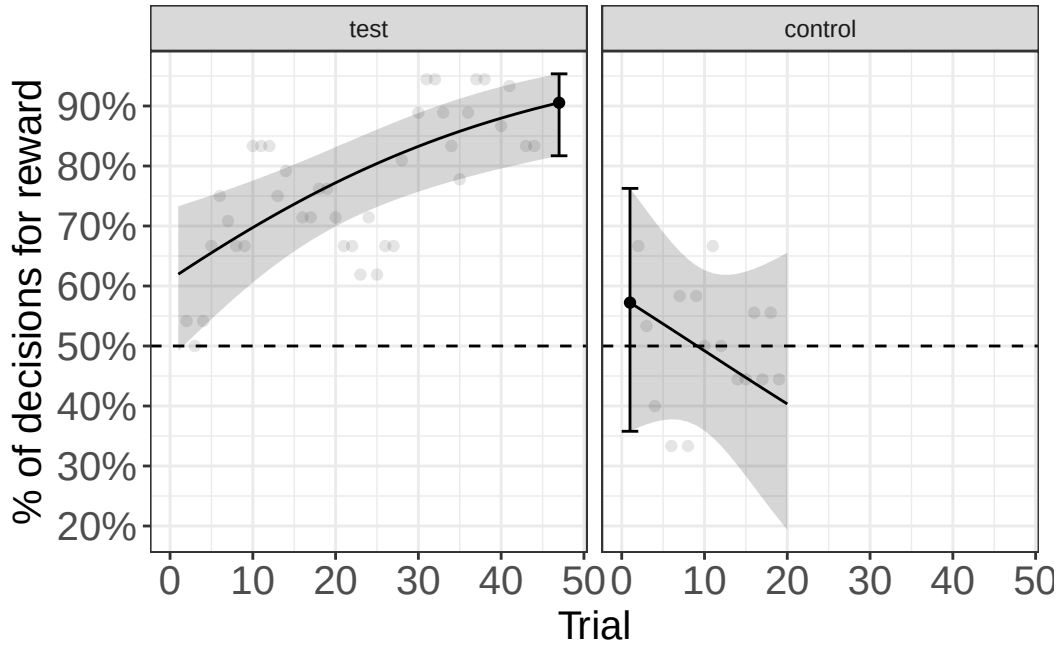


glm_analysis

Question 1: Can wasps discriminate a 'simple' artificial dorsal pattern?

Evidence that wasps can use overhead patterns comes from two observations. First, That the proportion of choices for the rewarding pattern is >50% and second, that this proportion of decisions for the rewarding patterns drops when the patterns are removed from the roof (during the control).



During the test phase of the experiment, trial number (experience) had a significant positive effect on the expected probability (P_{corr}) of a correct decision (Figure 1 solid line) with every successive trial increasing the log odds of a correct decision by 0.043 (95% CI [0.019, 0.067], $p < 0.001$).

The P_{corr} of a correct choice at the end of the test was $P_{\text{corr}} = 0.905$ (95% CI [0.817, 0.954]). The lower bound of the 95% confidence interval ([insert lower bound here]) was $>50\%$, implying that P_{corr} was significantly higher chance (50%) (black error bar in Figure 1 test panel does not overlap horizontal dashed line). P_{corr} at the end of the experiment was also significantly higher than at the start of the control ($P_{\text{corr}} = 0.572$, 95% CI [0.358, 0.763]), and this difference was significant (log odds ratio = 1.968, $p < 0.001$) (Figure 1, note lack of overlap of error bars between test and control panel).

Also note that P_{corr} is significantly above chance (grey shaded 95% confidence ribbon in Figure 1 does not overlap the 50% line) from an early trial number and remains that way until the end of the test. In contrast, the P_{corr} in the control phase is never significantly above chance for the duration of the control (95% confidence ribbon overlaps the 50% line), as one would expect for wasps making choices in the absence of any information regarding the location of the reward.

That the model suggests naive wasps perform above chance at the onset of the test phase (ie. in trial 1) can be attributed to the fact that the model is unable to fit the rapid learning of the wasps, rather than to the availability and learning of olfactory and / or temporal cues during the pre-test training phase. This is supported by the proportion of correct decisions in the first few 3-trial wide windows of the test phase (black points) that are clustered around 50%.

Question 2: What information are wasps using to discriminate patterns?

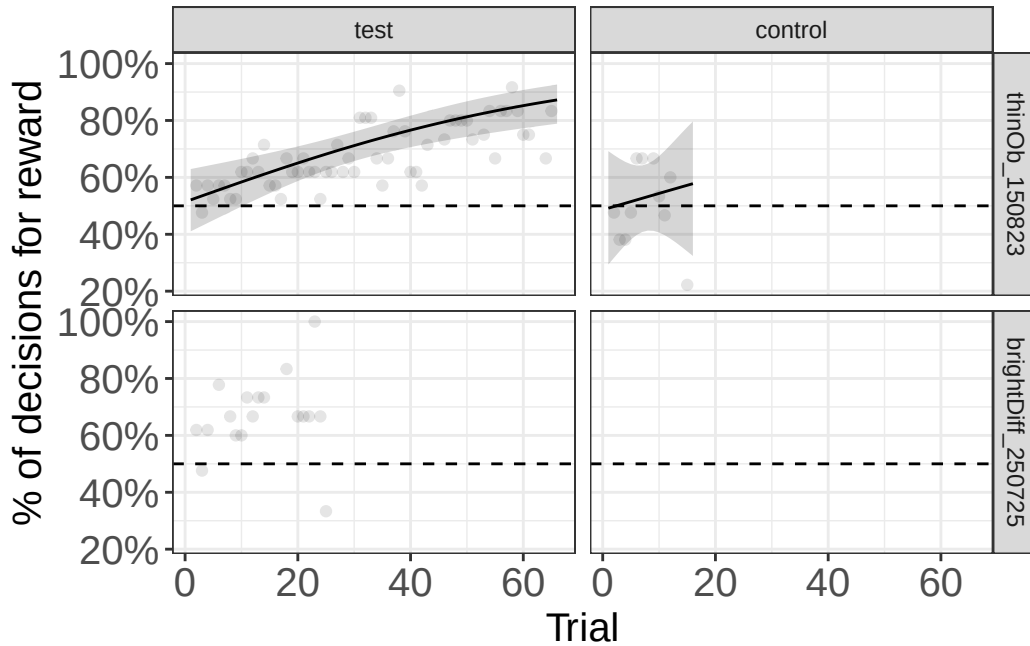


Table 1: Q2 trial-slope summary by artificial experiment

	logOdds	confint	pFmt	oddsMultiplier
brightDiff_250725	0.033	[-0.037, 0.103]	0.356	1.033
perpPara_170725	0.043	[0.019, 0.067]	<0.001	1.044
thinOb_150823	0.029	[0.015, 0.044]	<0.001	1.030

To answer whether wasps require optic flow information to discriminate patterns we tested whether they could distinguish a new artificial pattern-pair, one consisting of oblique stripes oriented at either 45 degrees to the right or to the left of the y-maze arm's long axis and the general direction of wasp travel. Here again, we tested whether the proportion of correct choices for this pattern-pair was >50% as evidence that wasps do not rely exclusively on optic flow to discriminate among stimuli. We did not run contr... (to do, incl. control data and use facet grid figure with only the bright diff and oblique data). also to do: add the test-control contrast table.

Wasps had no trouble distinguishing between oblique stripes that presumably produce identical rates of optic flow in the dorsal visual field (Figure 2). Similar to the perpendicular-parallel experiment, trial had a significant positive effect on P_{corr} , with every successive trial increasing

the log odds of a correct decision by 0.029 (95% CI [0.015, 0.044], $p < 0.001$; Table 1). Initially (at trial 1) performance was not significantly different from chance ($P_{\text{corr}} = 0.521$, 95% CI [0.409, 0.631]) but rose steeply, reaching 0.880 (95% CI [0.796, 0.932]) by the end of the experiment (trial 66), which is significantly higher than expected by chance (50%).

Next, we asked whether wasps could discriminate between an opaque and transparent roof, ie. between stimuli that carried little to no pattern information. While trial had a positive effect on P_{corr} (beta = 0.033, 95% CI [-0.037, 0.103]), unlike for the other two pairs of artificial stimuli this effect was not significant ($p = 0.356$), likely due to the much shorter number of trials this experiment was run for. While statistical support for a positive slope is lacking, there is support for $P_{\text{corr}} > 50\%$ towards the middle and end of the experiment. At the onset of the experiment (trial 1), wasp performance was not significantly different from 50% ($P_{\text{corr}} = 0.610$, 95% CI [0.436, 0.759]) but rises with experience and by the end of the experiment (trial 26) P_{corr} reached 0.780 (95% CI [0.500, 0.927]). Performance at the end of the experiment is only borderline significantly higher than 50% because of the increased uncertainty in the slope / effect of trials on performance due to the lower number of trials run in this experiment. However, if one uses other metrics or analyses to compare performance to the 50% line, they are significantly different (for example, comparing performance at an intermediate trial or comparing the average performance of the last five trials to the 50% line).

NBNB! Slope or brightdiff exp not significant. fade best fit line?

Comparison among different artificial stimuli: (to do - how does one turn this into an italics heading with line break?)

to do: - comparison of prop correct at the same trial number? - comparison of slopes? - What would be the appropriate way to compare among these experiments?

issues: - day is multicollinear in thin oblique experiment. It makes the slope of thin oblique appear less steep. This drives the difference in slopes detected between perppara and thin oblique. The issue is, is it real? I'm not confident enough that I ran the experiments in exactly the same way so as to make them comparable. If we remove day from the analysis, would we still find a significant difference in the slopes? Could the difference simply be attributable to slower learning due to the side bias which is clearly evident in the thin oblique experiment (possibly caused because of different pre-testing training regimes). Why even is there a side bias when these wasps had just come off the back of a previous experiment with a thick oblique stripe? was the side bias present at that time? - the issue im trying to get at above is that conclusions regarding slope diffs depend on the model you run. - massive confidence intervals around brightness experiment

Question 3: Can wasps use naturalistic patterns for navigation?

- Are performances different from 50%?
- Are performances different from the control?

- Are there differences in performance between naturalistic and artificial stimuli?

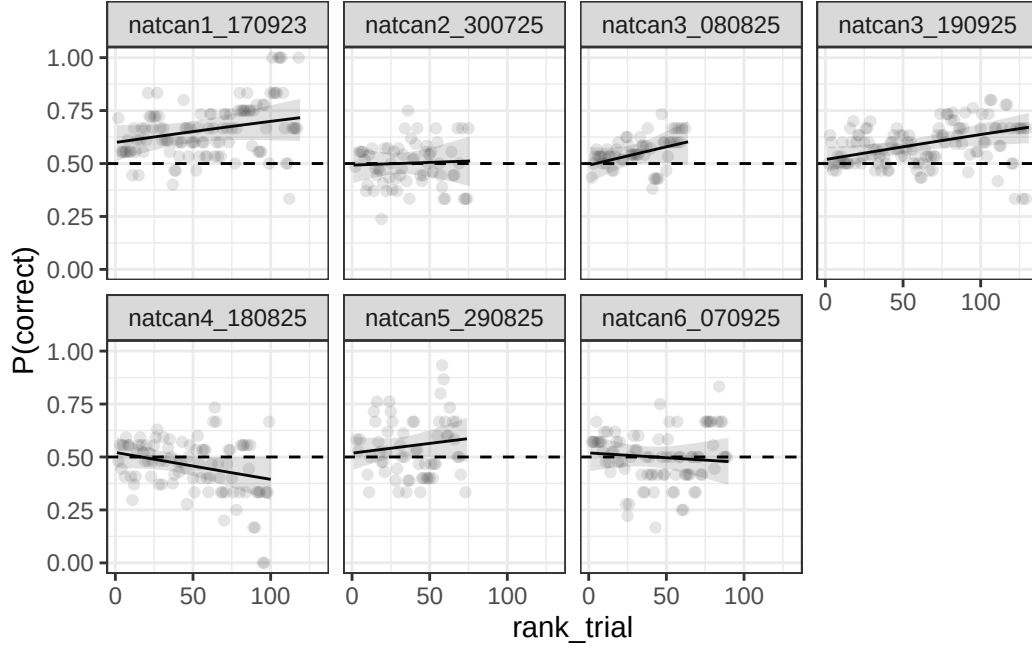


Table 2: Naturalistic test performance at first and last trial

experiment	nat_or_art	startPcorr95CI	endPcorr95CI	endTrial
natcan1_170923	naturalistic	0.600 [0.513, 0.682]	0.716 [0.606, 0.805]	119
natcan2_300725	naturalistic	0.493 [0.408, 0.577]	0.512 [0.394, 0.629]	76
natcan3_080825	naturalistic	0.494 [0.411, 0.577]	0.603 [0.495, 0.701]	64
natcan3_190925	naturalistic	0.520 [0.461, 0.578]	0.671 [0.595, 0.739]	131
natcan4_180825	naturalistic	0.520 [0.447, 0.593]	0.394 [0.296, 0.503]	100
natcan5_290825	naturalistic	0.519 [0.439, 0.598]	0.586 [0.479, 0.686]	74
natcan6_070925	naturalistic	0.519 [0.434, 0.603]	0.478 [0.368, 0.591]	90

Evidence for whether wasps can discriminate naturalistic canopies at above-chance rates ($>50\%$) is mixed. In four of the seven naturalistic canopy pairs we tested, the predicted P_{corr} by the end of the experiment was significantly above chance (the lower limit of the 95% CI is $> 50\%$, see Table x), suggesting wasps may be able to separate certain pairs of naturalistic canopies to some degree.

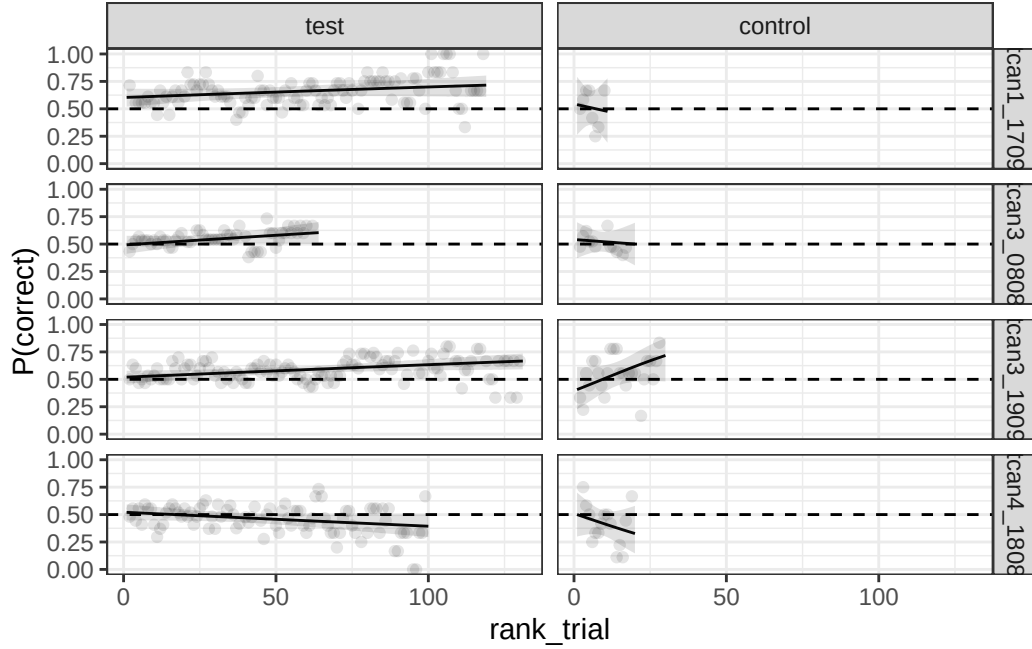


Table 3: Naturalistic test-control performance at test end and control start

experiment	testEndPcorr95CI	controlStartPcorr95CI	testEndTrial
natcan1_170923	0.716 [0.608, 0.803]	0.538 [0.263, 0.792]	119
natcan3_080825	0.604 [0.500, 0.700]	0.540 [0.372, 0.699]	64
natcan3_190925	0.667 [0.591, 0.735]	0.406 [0.227, 0.615]	131
natcan4_180825	0.394 [0.295, 0.502]	0.500 [0.300, 0.700]	100

Table 4: Naturalistic test-control contrasts (test end vs control start)

Experiment	Log-odds contrast	95% CI	P	Control start trial	Test end trial
natcan1_170923	3.771	[-0.494, 2.036]	0.232	1	119
natcan3_080825	0.265	[-0.538, 1.068]	0.518	1	64
natcan3_190925	1.072	[0.164, 1.980]	0.021	1	131
natcan4_180825	-0.431	[-1.381, 0.519]	0.374	1	100

naturalistic test vs control Of the four naturalistic canopies that show $P_{\text{corr}} > 50\%$, after three of those we also ran a control (patterns removed) directly after the test phase concluded, but in only one case (natcan1_170923) was the P_{corr} at the end of the test significantly higher than at the start of the control. This can be attributed to the fact that P_{corr} for the naturalistic canopies

is generally not much greater than 50% (if at all), and the fact that control experiments were not run for many trials, increasing the uncertainty around the P_{corr} during the control phase.

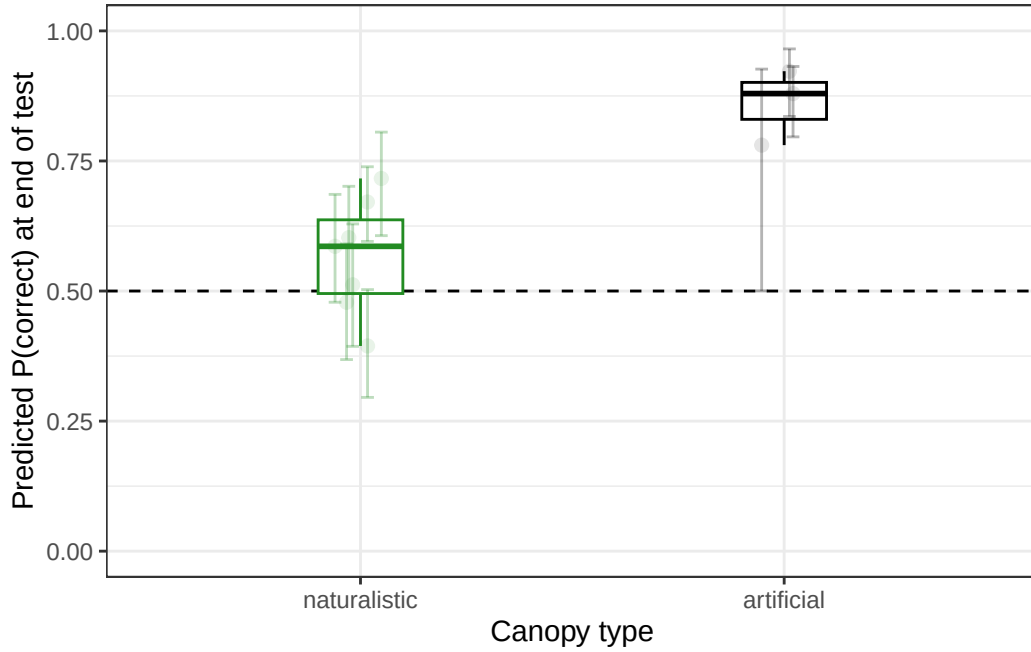


Figure 1: End-of-test predicted performance by canopy type. Each point is one experiment; vertical bars are 95% CIs.

nat vs art: Discrimination performance for naturalistic canopies tends to be lower than for artificial canopies. The naturalistic canopy showing the best performance (natcan1_170923, $P_{\text{corr}} = 0.716$) still has a lower P_{corr} than the artificial canopy with the worst performance (brightDiff_250725, $P_{\text{corr}} = 0.780$). The average performance for all naturalistic canopies is 0.566, whereas for artificial it is 0.861, and the difference in the average P_{corr} for naturalistic and artificial is 0.295. Given the low number of artificial and naturalistic experiments included here ($n = 3$ and 7 respectively) we did not test for an overall difference between naturalistic and artificial canopies.

ToDo:

Formatting - remove weird dash formatting under q1 - remove results at top of document - several places missing their statistics - change figures such that lines are in front of rolling mean points - perhasp change rolling mean points to become transparent - is the way i introduce a results section correct? - what is the logic behind when / where to reference tables and figures in line?

- Run an overall test comparing naturalistic and artificial canopies (including all data)

- merge testPTs table and Tc table for q3
- merge models, just use Tc model
- add end-trial estimates and CIs to all figures (q2 and q3 outstanding)
- x and y figure axis labels
- table headers and rownames need to be formatted for pub.
- pub-ready experiment names?
 - eg. perpendicular / parallel and oblique (+45 - 45)
- diffuser exp data in supplementary
- naturalistic -> artificial transition experiment

Supplementary

Table: Test predictions at start, middle, and end trials

experiment	nat_or_art	startPcorr95CI	midPcorr95CI	endPcorr95CI	m
:-----	:-----	:-----	:-----	:-----	:
brightDiff_250725	artificial	0.61 [0.44, 0.76]	0.70 [0.58, 0.81]	0.78 [0.50, 0.93]	1
natcan1_170923	naturalistic	0.60 [0.51, 0.68]	0.66 [0.60, 0.71]	0.72 [0.61, 0.81]	6
natcan2_300725	naturalistic	0.49 [0.41, 0.58]	0.50 [0.45, 0.56]	0.51 [0.39, 0.63]	3
natcan3_080825	naturalistic	0.49 [0.41, 0.58]	0.55 [0.50, 0.60]	0.60 [0.49, 0.70]	3
natcan3_190925	naturalistic	0.52 [0.46, 0.58]	0.60 [0.56, 0.63]	0.67 [0.59, 0.74]	6
natcan4_180825	naturalistic	0.52 [0.45, 0.59]	0.46 [0.41, 0.50]	0.39 [0.30, 0.50]	5
natcan5_290825	naturalistic	0.52 [0.44, 0.60]	0.55 [0.50, 0.60]	0.59 [0.48, 0.69]	3
natcan6_070925	naturalistic	0.52 [0.43, 0.60]	0.50 [0.45, 0.55]	0.48 [0.37, 0.59]	4
perpPara_170725	artificial	0.62 [0.48, 0.75]	0.82 [0.73, 0.88]	0.92 [0.84, 0.96]	2
thinOb_150823	artificial	0.52 [0.41, 0.63]	0.74 [0.68, 0.79]	0.88 [0.80, 0.93]	3

Table: Test-control predictions at test end and control start

experiment	nat_or_art	controlStartPcorr95CI	testEndPcorr95CI	testEndTrial
:-----	:-----	:-----	:-----	-----:
natcan1_170923	naturalistic	0.538 [0.263, 0.792]	0.716 [0.608, 0.803]	119
natcan3_080825	naturalistic	0.540 [0.372, 0.699]	0.604 [0.500, 0.700]	64
natcan3_190925	naturalistic	0.406 [0.227, 0.615]	0.667 [0.591, 0.735]	131
natcan4_180825	naturalistic	0.500 [0.300, 0.700]	0.394 [0.295, 0.502]	100

perpPara_170725	artificial	0.572 [0.358, 0.763]	0.905 [0.817, 0.954]		47
thinOb_150823	artificial	0.492 [0.294, 0.692]	0.872 [0.789, 0.926]		66

Table: Test-control contrast summary

experiment	logOddsRatio	confint	pFmt	controlStartTrial	testEndTrial
:-----	:-----	:-----	:-----	:-----	:-----
natcan1_170923	0.771	[-0.494, 2.036]	0.232	1	119
natcan3_080825	0.265	[-0.538, 1.068]	0.518	1	64
natcan3_190925	1.072	[0.164, 1.980]	0.021	1	131
natcan4_180825	-0.431	[-1.381, 0.519]	0.374	1	100
perpPara_170725	1.968	[0.881, 3.055]	<0.001	1	47
thinOb_150823	1.954	[0.907, 3.002]	<0.001	1	66

Table: Test trial-slope summary

experiment	nat_or_art	logOdds	confint	pFmt	oddsMultiplier
:-----	:-----	:-----	:-----	:-----	:-----
brightDiff_250725	artificial	0.033	[-0.037, 0.103]	0.356	1.033
perpPara_170725	artificial	0.043	[0.019, 0.067]	<0.001	1.044
thinOb_150823	artificial	0.029	[0.015, 0.044]	<0.001	1.030
natcan1_170923	naturalistic	0.004	[-0.002, 0.010]	0.153	1.004
natcan2_300725	naturalistic	0.001	[-0.008, 0.010]	0.832	1.001
natcan3_080825	naturalistic	0.007	[-0.004, 0.018]	0.195	1.007
natcan3_190925	naturalistic	0.005	[0.001, 0.009]	0.013	1.005
natcan4_180825	naturalistic	-0.005	[-0.012, 0.001]	0.122	0.995
natcan5_290825	naturalistic	0.004	[-0.005, 0.013]	0.412	1.004
natcan6_070925	naturalistic	-0.002	[-0.010, 0.006]	0.647	0.998

Table: Test-control trial-slope summary

experiment	nat_or_art	logOdds	confint	pFmt	oddsMultiplier
:-----	:-----	:-----	:-----	:-----	:-----
brightDiff_250725	artificial	0.033	[-0.037, 0.103]	0.356	1.033

perpPara_170725	artificial	0.043	[0.019, 0.067]	<0.001	1.044	
thin0b_150823	artificial	0.029	[0.015, 0.044]	<0.001	1.030	
natcan1_170923	naturalistic	0.004	[-0.002, 0.010]	0.153	1.004	
natcan2_300725	naturalistic	0.001	[-0.008, 0.010]	0.832	1.001	
natcan3_080825	naturalistic	0.007	[-0.004, 0.018]	0.195	1.007	
natcan3_190925	naturalistic	0.005	[0.001, 0.009]	0.013	1.005	
natcan4_180825	naturalistic	-0.005	[-0.012, 0.001]	0.122	0.995	
natcan5_290825	naturalistic	0.004	[-0.005, 0.013]	0.412	1.004	
natcan6_070925	naturalistic	-0.002	[-0.010, 0.006]	0.647	0.998	

Table: Test model coefficients

experiment	nat_or_art	term	logOdds	confint	pFmt	
:-----	:-----	:-----	:-----	:-----	:-----	
brightDiff_250725	artificial	(Intercept)	1.049	[0.101, 1.997]	0.030	
brightDiff_250725	artificial	rank_trial	0.033	[-0.037, 0.103]	0.356	
brightDiff_250725	artificial	sideAlt_trialTRUE	-0.381	[-1.212, 0.450]	0.369	
brightDiff_250725	artificial	reward_sideR	-0.892	[-1.760, -0.024]	0.044	
brightDiff_250725	artificial	sd__(Intercept)	0.000	[NA, NA]	NA	
perpPara_170725	artificial	(Intercept)	1.130	[0.385, 1.876]	0.003	
perpPara_170725	artificial	rank_trial	0.043	[0.019, 0.067]	<0.001	
perpPara_170725	artificial	sideAlt_trialTRUE	0.141	[-0.438, 0.721]	0.633	
perpPara_170725	artificial	reward_sideR	-1.479	[-2.070, -0.888]	<0.001	
perpPara_170725	artificial	sd__(Intercept)	0.496	[NA, NA]	NA	
thin0b_150823	artificial	(Intercept)	-1.122	[-1.709, -0.535]	<0.001	
thin0b_150823	artificial	rank_trial	0.029	[0.015, 0.044]	<0.001	
thin0b_150823	artificial	sideAlt_trialTRUE	0.021	[-0.464, 0.505]	0.933	
thin0b_150823	artificial	reward_sideR	2.330	[1.801, 2.860]	<0.001	
thin0b_150823	artificial	sd__(Intercept)	0.000	[NA, NA]	NA	
natcan1_170923	naturalistic	(Intercept)	-0.090	[-0.520, 0.341]	0.683	
natcan1_170923	naturalistic	rank_trial	0.004	[-0.002, 0.010]	0.153	
natcan1_170923	naturalistic	sideAlt_trialTRUE	0.204	[-0.161, 0.569]	0.273	
natcan1_170923	naturalistic	reward_sideR	0.781	[0.416, 1.146]	<0.001	
natcan1_170923	naturalistic	sd__(Intercept)	0.159	[NA, NA]	NA	
natcan2_300725	naturalistic	(Intercept)	-0.524	[-0.930, -0.118]	0.011	
natcan2_300725	naturalistic	rank_trial	0.001	[-0.008, 0.010]	0.832	
natcan2_300725	naturalistic	sideAlt_trialTRUE	0.048	[-0.315, 0.412]	0.794	
natcan2_300725	naturalistic	reward_sideR	0.940	[0.581, 1.299]	<0.001	
natcan2_300725	naturalistic	sd__(Intercept)	0.122	[NA, NA]	NA	
natcan3_080825	naturalistic	(Intercept)	0.388	[-0.031, 0.808]	0.070	

natcan3_080825	naturalistic	rank_trial	0.007	[-0.004, 0.018]	0.195	
natcan3_080825	naturalistic	sideAlt_trialTRUE	-0.211	[-0.573, 0.152]	0.255	
natcan3_080825	naturalistic	reward_sideR	-0.629	[-0.992, -0.266]	<0.001	
natcan3_080825	naturalistic	sd__(Intercept)	0.089	[NA, NA]	NA	
natcan3_190925	naturalistic	(Intercept)	-0.036	[-0.334, 0.262]	0.812	
natcan3_190925	naturalistic	rank_trial	0.005	[0.001, 0.009]	0.013	
natcan3_190925	naturalistic	sideAlt_trialTRUE	-0.034	[-0.287, 0.220]	0.795	
natcan3_190925	naturalistic	reward_sideR	0.256	[0.010, 0.502]	0.042	
natcan3_190925	naturalistic	sd__(Intercept)	0.000	[NA, NA]	NA	
natcan4_180825	naturalistic	(Intercept)	-0.114	[-0.474, 0.245]	0.533	
natcan4_180825	naturalistic	rank_trial	-0.005	[-0.012, 0.001]	0.122	
natcan4_180825	naturalistic	sideAlt_trialTRUE	-0.207	[-0.535, 0.121]	0.216	
natcan4_180825	naturalistic	reward_sideR	0.608	[0.283, 0.933]	<0.001	
natcan4_180825	naturalistic	sd__(Intercept)	0.076	[NA, NA]	NA	
natcan5_290825	naturalistic	(Intercept)	0.597	[0.155, 1.038]	0.008	
natcan5_290825	naturalistic	rank_trial	0.004	[-0.005, 0.013]	0.412	
natcan5_290825	naturalistic	sideAlt_trialTRUE	-0.257	[-0.632, 0.118]	0.179	
natcan5_290825	naturalistic	reward_sideR	-0.792	[-1.168, -0.415]	<0.001	
natcan5_290825	naturalistic	sd__(Intercept)	0.000	[NA, NA]	NA	
natcan6_070925	naturalistic	(Intercept)	0.237	[-0.194, 0.667]	0.281	
natcan6_070925	naturalistic	rank_trial	-0.002	[-0.010, 0.006]	0.647	
natcan6_070925	naturalistic	sideAlt_trialTRUE	-0.168	[-0.559, 0.222]	0.398	
natcan6_070925	naturalistic	reward_sideR	-0.151	[-0.536, 0.234]	0.442	
natcan6_070925	naturalistic	sd__(Intercept)	0.000	[NA, NA]	NA	

Table: Test-control model coefficients

experiment	nat_or_art	term	logOdds	confint	pFmt	
:-----	:-----	:-----	:-----	:-----	:-----	
brightDiff_250725	artificial	(Intercept)	1.049	[0.101, 1.997]	0.030	
brightDiff_250725	artificial	rank_trial	0.033	[-0.037, 0.103]	0.356	
brightDiff_250725	artificial	sideAlt_trialTRUE	-0.381	[-1.212, 0.450]	0.369	
brightDiff_250725	artificial	reward_sideR	-0.892	[-1.760, -0.024]	0.044	
brightDiff_250725	artificial	sd__(Intercept)	0.000	[NA, NA]	NA	
perpPara_170725	artificial	(Intercept)	1.130	[0.385, 1.876]	0.003	
perpPara_170725	artificial	rank_trial	0.043	[0.019, 0.067]	<0.001	
perpPara_170725	artificial	sideAlt_trialTRUE	0.141	[-0.438, 0.721]	0.633	
perpPara_170725	artificial	reward_sideR	-1.479	[-2.070, -0.888]	<0.001	
perpPara_170725	artificial	sd__(Intercept)	0.496	[NA, NA]	NA	
thinOb_150823	artificial	(Intercept)	-1.122	[-1.709, -0.535]	<0.001	

thin0b_150823	artificial	rank_trial	0.029	[0.015, 0.044]	<0.001
thin0b_150823	artificial	sideAlt_trialTRUE	0.021	[-0.464, 0.505]	0.933
thin0b_150823	artificial	reward_sideR	2.330	[1.801, 2.860]	<0.001
thin0b_150823	artificial	sd__(Intercept)	0.000	[NA, NA]	NA
natcan1_170923	naturalistic	(Intercept)	-0.090	[-0.520, 0.341]	0.683
natcan1_170923	naturalistic	rank_trial	0.004	[-0.002, 0.010]	0.153
natcan1_170923	naturalistic	sideAlt_trialTRUE	0.204	[-0.161, 0.569]	0.273
natcan1_170923	naturalistic	reward_sideR	0.781	[0.416, 1.146]	<0.001
natcan1_170923	naturalistic	sd__(Intercept)	0.159	[NA, NA]	NA
natcan2_300725	naturalistic	(Intercept)	-0.524	[-0.930, -0.118]	0.011
natcan2_300725	naturalistic	rank_trial	0.001	[-0.008, 0.010]	0.832
natcan2_300725	naturalistic	sideAlt_trialTRUE	0.048	[-0.315, 0.412]	0.794
natcan2_300725	naturalistic	reward_sideR	0.940	[0.581, 1.299]	<0.001
natcan2_300725	naturalistic	sd__(Intercept)	0.122	[NA, NA]	NA
natcan3_080825	naturalistic	(Intercept)	0.388	[-0.031, 0.808]	0.070
natcan3_080825	naturalistic	rank_trial	0.007	[-0.004, 0.018]	0.195
natcan3_080825	naturalistic	sideAlt_trialTRUE	-0.211	[-0.573, 0.152]	0.255
natcan3_080825	naturalistic	reward_sideR	-0.629	[-0.992, -0.266]	<0.001
natcan3_080825	naturalistic	sd__(Intercept)	0.089	[NA, NA]	NA
natcan3_190925	naturalistic	(Intercept)	-0.036	[-0.334, 0.262]	0.812
natcan3_190925	naturalistic	rank_trial	0.005	[0.001, 0.009]	0.013
natcan3_190925	naturalistic	sideAlt_trialTRUE	-0.034	[-0.287, 0.220]	0.795
natcan3_190925	naturalistic	reward_sideR	0.256	[0.010, 0.502]	0.042
natcan3_190925	naturalistic	sd__(Intercept)	0.000	[NA, NA]	NA
natcan4_180825	naturalistic	(Intercept)	-0.114	[-0.474, 0.245]	0.533
natcan4_180825	naturalistic	rank_trial	-0.005	[-0.012, 0.001]	0.122
natcan4_180825	naturalistic	sideAlt_trialTRUE	-0.207	[-0.535, 0.121]	0.216
natcan4_180825	naturalistic	reward_sideR	0.608	[0.283, 0.933]	<0.001
natcan4_180825	naturalistic	sd__(Intercept)	0.076	[NA, NA]	NA
natcan5_290825	naturalistic	(Intercept)	0.597	[0.155, 1.038]	0.008
natcan5_290825	naturalistic	rank_trial	0.004	[-0.005, 0.013]	0.412
natcan5_290825	naturalistic	sideAlt_trialTRUE	-0.257	[-0.632, 0.118]	0.179
natcan5_290825	naturalistic	reward_sideR	-0.792	[-1.168, -0.415]	<0.001
natcan5_290825	naturalistic	sd__(Intercept)	0.000	[NA, NA]	NA
natcan6_070925	naturalistic	(Intercept)	0.237	[-0.194, 0.667]	0.281
natcan6_070925	naturalistic	rank_trial	-0.002	[-0.010, 0.006]	0.647
natcan6_070925	naturalistic	sideAlt_trialTRUE	-0.168	[-0.559, 0.222]	0.398
natcan6_070925	naturalistic	reward_sideR	-0.151	[-0.536, 0.234]	0.442
natcan6_070925	naturalistic	sd__(Intercept)	0.000	[NA, NA]	NA